

1. Scan for available Wi-Fi networks on your laptop or phone and list the SSIDs you find, noting which ones are open (unsecured) and which require a password.

No.	SSID (Wi-Fi Name)	Security
1	Manpri	Secured – Password required
2	Rohan's Samsung galaxy s22 plus	Secured – Password required
3	Vrajbhoomi_Library	Open – No Password
4	Jiofiber	Secured – Password required

2. Set up a mobile hotspot on your phone, connect a laptop to it, and check what IP address your laptop receives from the hotspot.

When I check IP address with my home Wi-Fi

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Wireless LAN adapter WiFi:

Connection-specific DNS Suffix  . : 
IPv6 Address. . . . . : 2001:db8:1::100
IPv6 Address. . . . . : 2402:a00:408:8f67:7ec3:d8aa:6308:d824
IPv6 Address. . . . . : fd00:1:1::100
Temporary IPv6 Address. . . . . : 2402:a00:408:8f67:d159:6eb:c46b:af37
Link-local IPv6 Address . . . . . : fe80::bcff:c2db:3220:39b7%6
IPv4 Address. . . . . : 192.168.1.72
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : fe80::1%6
                             192.168.1.254
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When I check IP address with my mobile hotspot's Wi-Fi

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Wireless LAN adapter WiFi:

Connection-specific DNS Suffix  . : 
IPv6 Address. . . . . : 2001:db8:1::100
IPv6 Address. . . . . : fd00:1:1::100
Link-local IPv6 Address . . . . . : fe80::bcff:c2db:3220:39b7%6
IPv4 Address. . . . . : 10.57.208.75
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : 10.57.208.41
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3. Research and explain in your own words what WPA2 and WPA3 are, and list two differences between them relevant to security.

WPA2 and WPA3 are security standards used to protect Wi-Fi networks. They encrypt the data travelling between devices and the Wi-Fi router to help prevent unauthorized access and eavesdropping.

Point	WPA2	WPA3
1. Authentication	Commonly uses WPA2-Personal with a pre-shared key. Weak passwords can be vulnerable to password-guessing attacks.	Uses SAE which provides stronger protection against password-guessing and offline dictionary attacks.
2. Security Strength	Uses AES encryption providing strong security but the overall protection depends more heavily on the strength of the Wi-Fi password.	Provides stronger authentication and additional security protections making it harder for attackers to crack Wi-Fi credentials.

4. Find a news article or real-world case from the past 3 years where a public Wi-Fi network was used in a cyber-attack, and summarize how the attack happened and what could have prevented it.

Real-world case: Fake Wi-Fi networks in Australia – 2024

- In June 2024, Australian police charged a 42-year old man who allegedly created fake free Wi-Fi networks at several Australian airports and on domestic flights. The fake networks were designed to look like legitimate airport or airport Wi-Fi.

How the attack happened:

- The attacker created Wi-Fi networks with names that looked legitimate.
- Passengers searching for free airport/airline Wi-Fi could mistakenly connect to the attacker's network.

- After connecting users were directed to a fake web page asking for personal information such as email addresses or other login details.
- The attacker could collect information entered by unsuspecting users.

What could have prevented it?

- Verify the Wi-Fi name – Passengers should confirm the exact SSID with airport or airline staff rather than choosing a similarly named network.
- Use a VPN – A trusted VPN can encrypt internet traffic when using an untrusted network.
- Avoid automatic Wi-Fi connection – Disable automatic connection unfamiliar networks.
- Multi-factor Authentication – Even if a password is stolen, MFA can make it much harder for an attacker to access the account.